```
        if(flag==1)
                cout<<"\n The String is palindrome";
        else
                cout<<"\n The String is not palindrome";
        return 0;
}
```

Output:

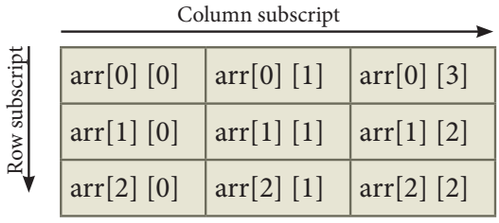
```
        Enter a string : madam
        The String is palindrome
```

12.3 Two-dimensional array

Two-dimensional (2D) arrays are collection of similar elements where the elements are stored in certain number of rows and columns. An example m × n matrix where m denotes the number of rows and n denotes the number of columns is shown in Figure12.4

int arr[3][3];

2D array conceptual memory representation



The array arr can be coneptually viewed in matrix form with 3 rows and 3 coloumns. The point to be noted here is since the subscript starts with 0 arr [0][0] represents the first element.

Figure 12.4

12.3.1 Declaration of 2-D array

The declaration of a 2-D array is

data-type array_name[row-size][col-size];

In the above declaration, data-type refers to any valid C++ data-type, array_name refers to the name of the 2-D array, row-size refers to the number of rows and col-size refers to the number of columns in the 2-D array.

For example

```
int A[3][4];
```

In the above example, A is a 2-D array, 3 denotes the number of rows and 4 denotes the number of columns. This array can hold a maximum of 12 elements.

Note

Array size must be an unsigned integer value which is greater than 0. In arrays, column size is compulsory but row size is optional.

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